



793D FINAL DRIVE PLANETARY GROUP LIFTING TOOL

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC Management

793D FINAL DRIVE PLANETARY GROUP LIFTING TOOL

1.0

Introduction

1

2.0

Best Practice Description

1

3.0

Implementation Steps.....

2

4.0

Benefits

3

5.0

Resources Required.....

3

6.0

Supporting Attachments / References.....

4

7.0

Related Best Practices

9

8.0

Acknowledgements

9

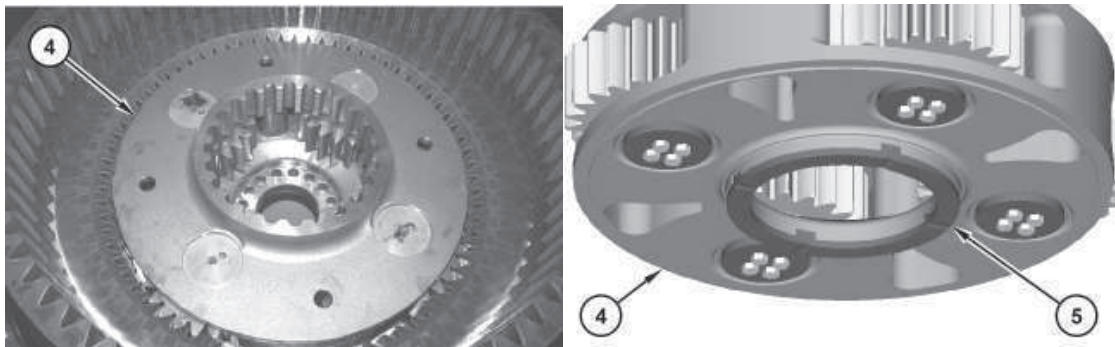
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1.0 Introduction

This best practice aims to facilitate and expedite the removal process for the planetary carrier in a 793D final drive through the design of a tool easily manufactured, in addition to being safe in its implementation and use.

During the component disassembly process, a series of tools and techniques are used according to the situation. In the case of a 793D truck final drive disassembly process, specifically the removal of the planetary carrier, Caterpillar does not indicate specifically how to perform this task; however, it is still an important part of the component disassembly process.

The idea for this best practice comes from the need to reduce the execution time of the work, and beyond that, to make it easy for the technician when conducting this task by using a tool that provides safety as the planetary carrier removal is performed, all this with the aim to improve the efficiency of the work.



2.0 Best Practice Description

The tool consists of two ASTM A36 steel sheets welded forming a "T". The first and shortest is 50 x 60, @ 3/4" which has a 15 mm hole where the lifting shackle is installed. The second and longest is 50 .8 x 368.3 @ 1" which has at one side a slot that allows equalization by means of a bolt (5/8 "- 18, 5" long) of one of the hook-shaped plates located at its ends. The plate of the opposite end is fitted through a similar bolt on a fixed hole that only allows rotation on its same axle; this allows the carrier planetarium clamping to be made from on the inside.

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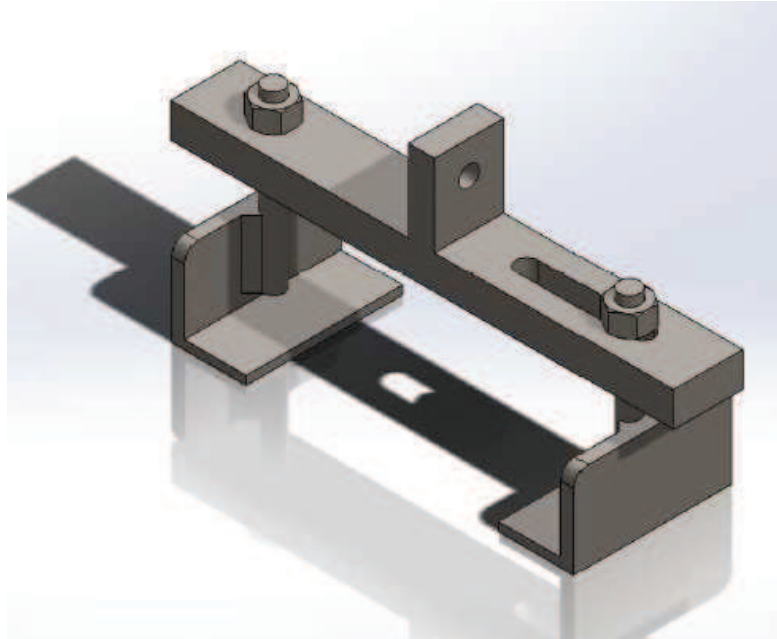
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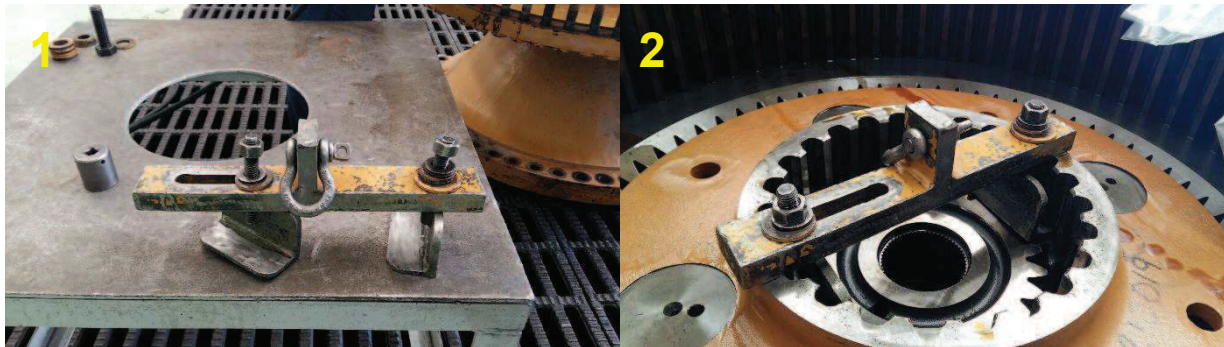
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Page 1 of 9



3.0 Implementation Steps

- Check the status of the tool (1) to verify the integrity of it before it's used.
- Once the final drive outside carrier is removed, place the lifting tool in the upper opening of the planetary, adjusted according to the diameter (2).
- Lift the planetary by means of the jib crane (3-4) to the desired height.
- Place the component in its final housing (pallet, basket, box) and remove the tools (5-6).



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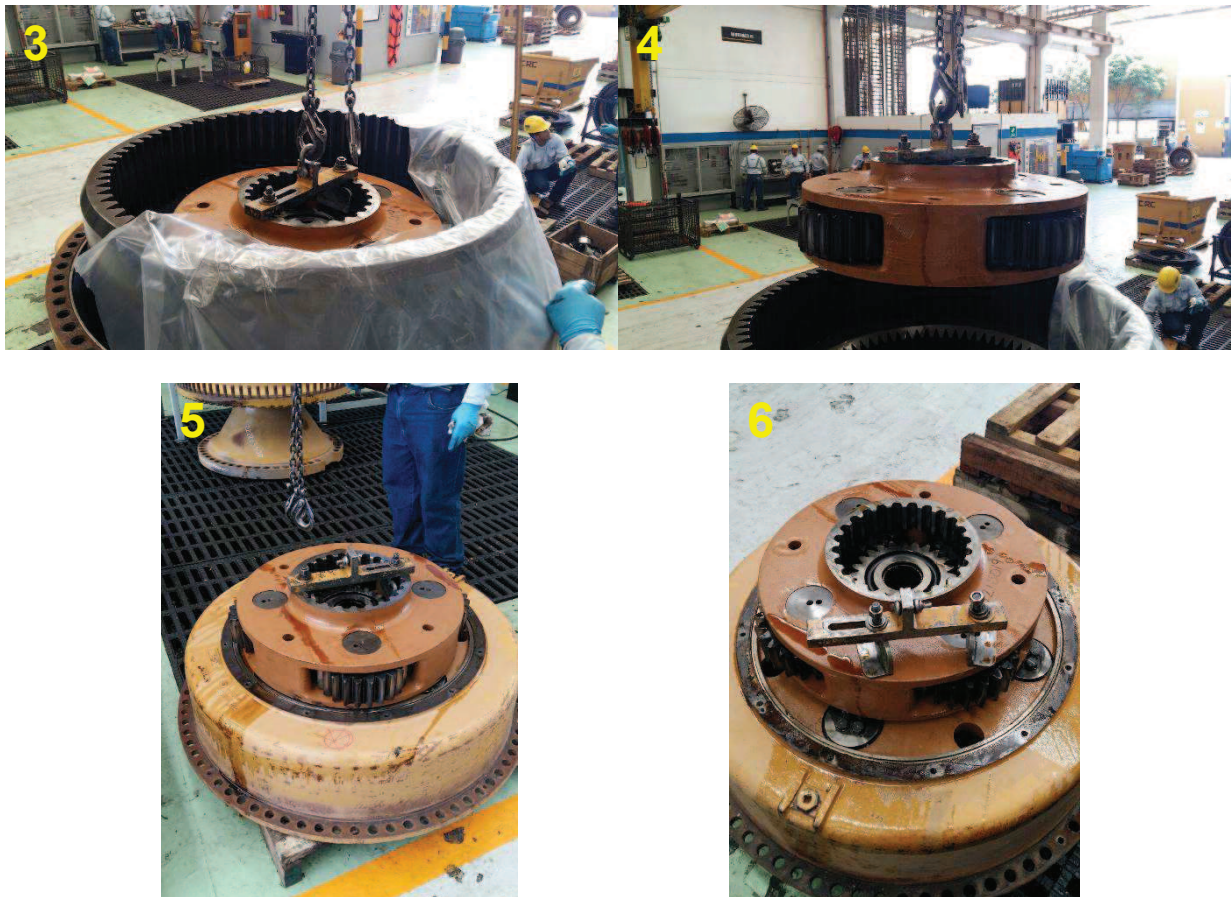
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4.0 Benefits

- The main benefits provided by this implementation are:
- Reduction in the planetary carrier removal time.
 - Ergonomic risk reduction to the technicians.
 - Easy to perform the work.

5.0 Resources Required

Welder, equipment and materials according to the tool drawings.

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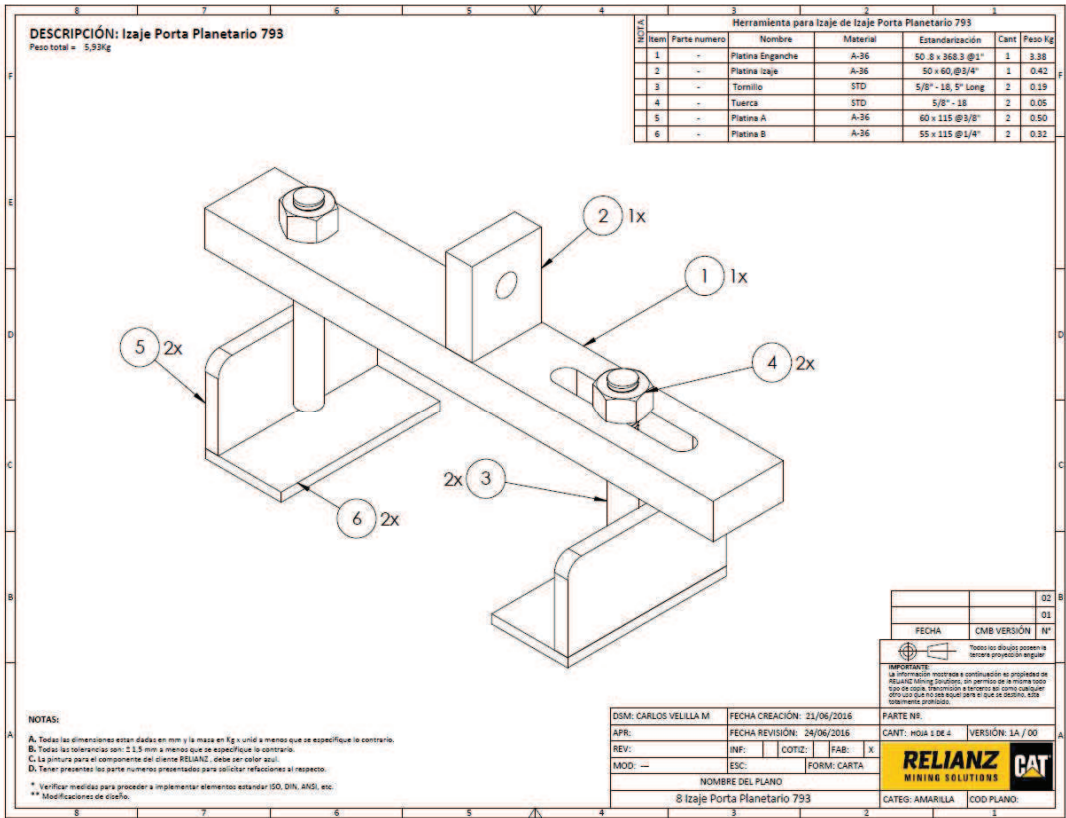
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6.0 Supporting Attachments / References



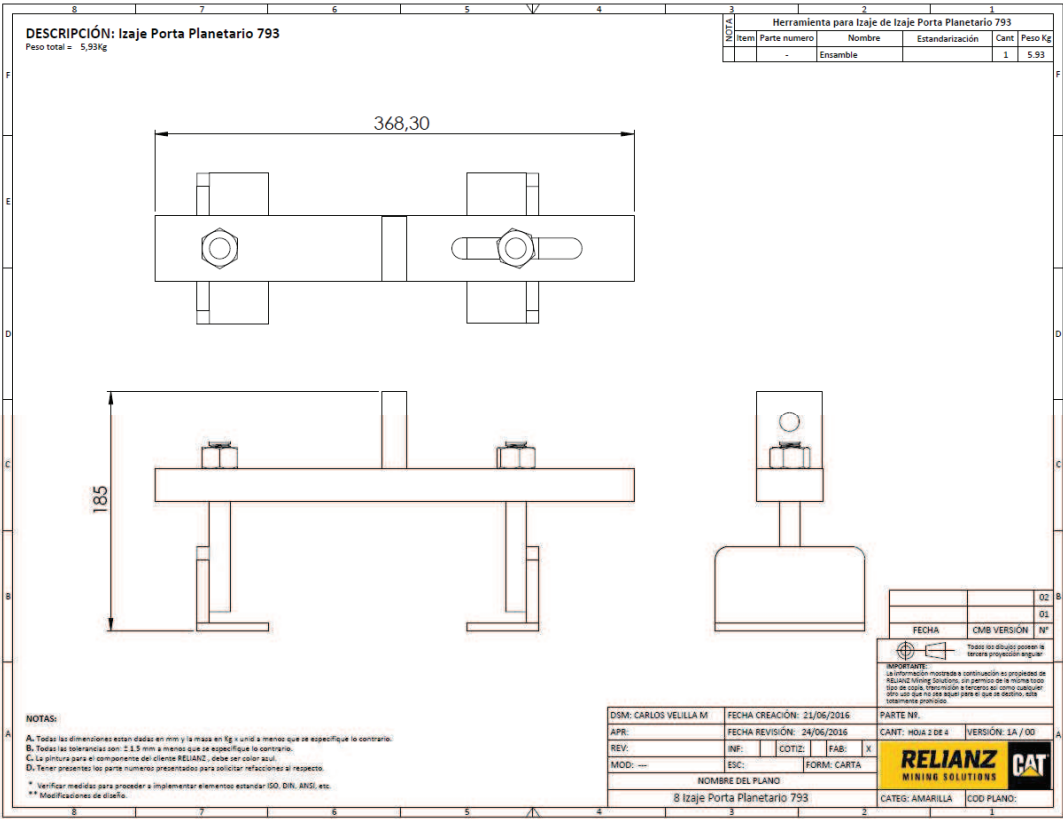
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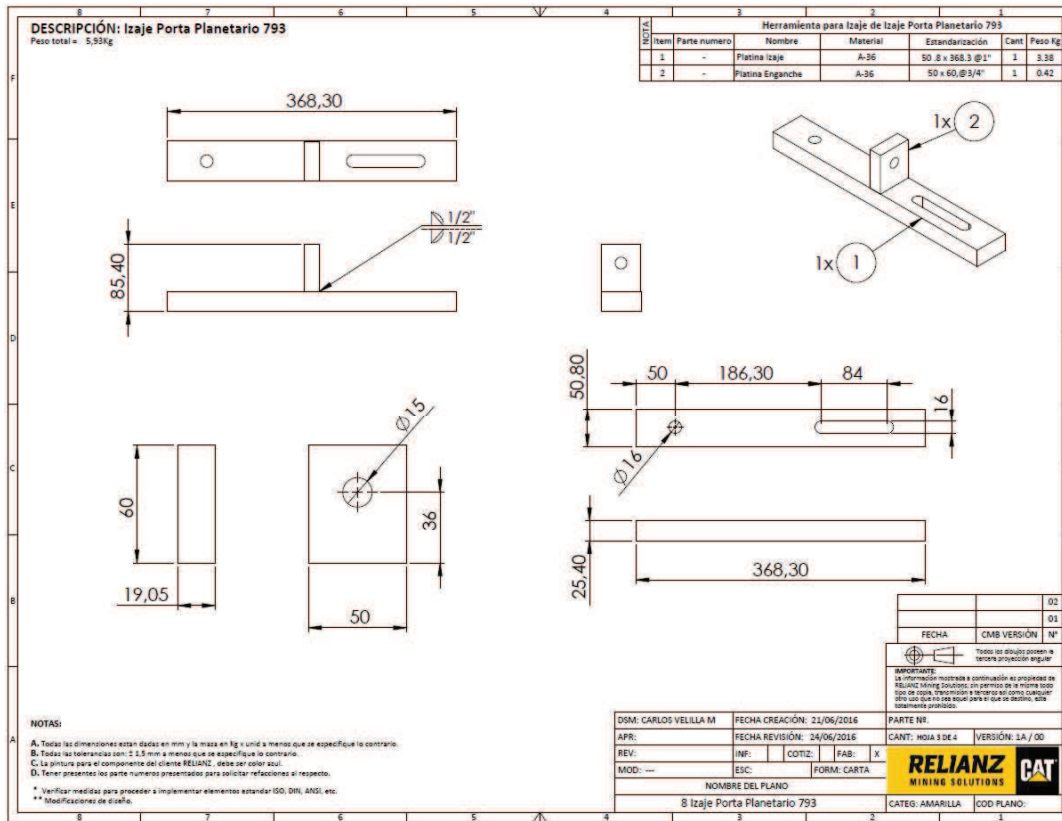
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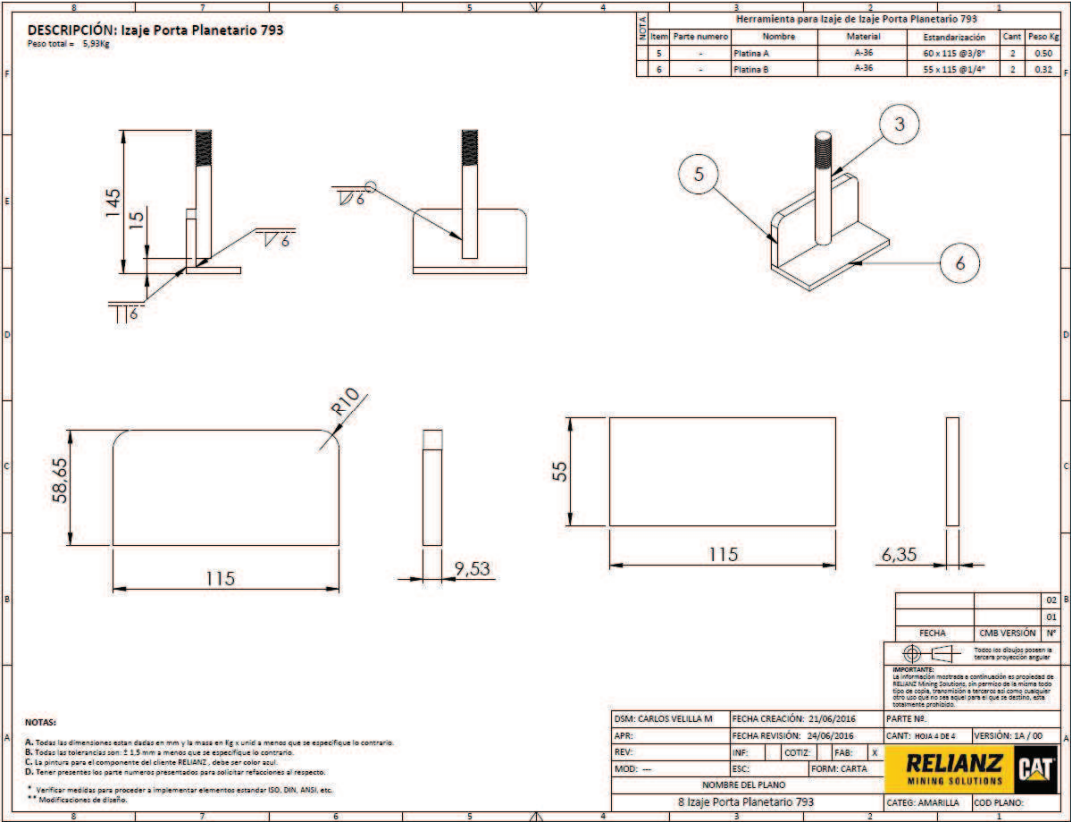
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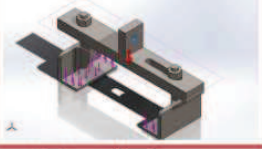
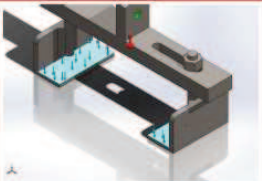
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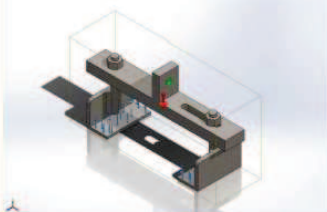
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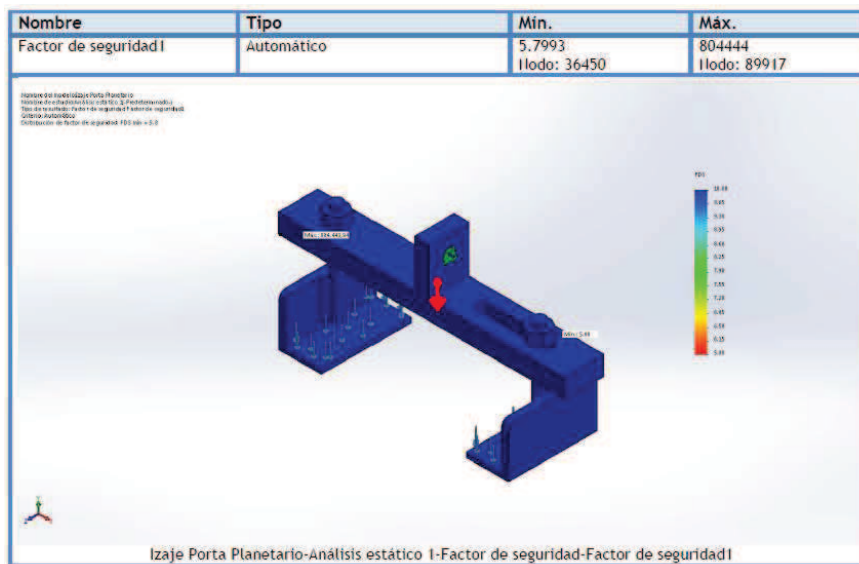
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Nombre de carga	Cargar imagen	Detalles de carga
Gravedad-1		Referencia: Planta Valores: 0 -9,81 Unidades: SI
Fuerza-1		Entidades: 2 cara(s) Tipo: Aplicar fuerza normal Valor: 1280 II

Información de contacto

Contacto	Imagen del contacto	Propiedades del contacto
Contacto global		Tipo: Unión rígida Componentes: 1 componente(s) Opciones: Mallado compatible



The minimum safety factor is 5.8, the maximum stress concentration occurs in the anchor point between the bolt and the supporting plate. However, since it is a safety factor greater than 5.0, it becomes a safe tool to be used.

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7.0 Related Best Practices

None

8.0 Acknowledgements

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